

AI Use Case Canvas

Water Potability Prediction API | Jan Pawel Bielecki (114012111) | Victoria Kammerer (114012110) | Winslet Clarisse (114012120)
AI Development and Security | Dataset: Kadiwal, A. (2021). Water Potability Dataset. Kaggle.
GitHub: <https://github.com/bielejan09/WaterPotabilityNN> | Live API: <http://72.155.73.40:5000>

USE CASE	RISK TIER	SYSTEM SCOPE	SDG ALIGNMENT
Predict drinking water safety from 9 physicochemical measurements using a PyTorch MLP Neural Network and Random Forest classifier. Decision support tool for field screening of water samples.	Medium-High Public health decision support. Misclassification of unsafe water as safe carries direct health consequences. Human review required before any action.	Public REST API on Azure VM. HTML web interface and JSON endpoint. Two models deployed simultaneously with dynamic recommendation.	SDG 6: Clean Water and Sanitation Supports access to safe drinking water monitoring in resource-limited settings where laboratory testing is expensive or unavailable.
ML MODELS	PERFORMANCE TARGETS	HUMAN OVERSIGHT	DEPLOYMENT
Primary: PyTorch MLP v1 Threshold: 0.7 Recall (unsafe): 95.2% False negatives: 19/656 Secondary: Random Forest Threshold: 0.6 Recall (unsafe): 93.2% False negatives: 27/656	Minimum recall on non-potable class: 0.90 Maximum false negatives per 656 samples: 30 Both models must agree on potable result before high confidence flag.	Human review required before acting on any prediction. Model is decision support only, not a final safety certificate. Field worker must verify borderline results with laboratory testing.	Platform: Azure VM IP: 72.155.73.40:5000 Framework: Flask Language: Python 3.12 Status: Live Monitoring: predictions.log Security: Rate limiting, input validation, headers
MLSECOPS CONTROLS		KNOWN LIMITATIONS	
Data Card and Model Card documented Leakage check: all correlations below 0.1 SMOTE applied to training set only StandardScaler fitted on training data only Threshold tuned to maximise recall on unsafe class Input validation: 9 features, non-negative, pH 0-14 Rate limiting: 10 requests per IP per minute Security headers on all API responses Request size limit: 16 KB Prediction logging with timestamps and IP addresses End of life plan documented in model card		No biological contamination detection, bacterial and viral threats are invisible to physicochemical measurements Geographic origin of training data undocumented: generalisation to other regions unvalidated Missing values in pH (14.99%) and Sulfate (23.84%) filled by median imputation Static model : no retraining mechanism in current deployment HTTP only : no SSL encryption in course project deployment Model drift monitoring not automated : manual review only	
TEVV EVIDENCE		SUCCESS CRITERIA	
Performance: Accuracy 67.7%, Recall 95.2%, F1 0.782, confusion matrices for all 5 models, Fairness: No protected attributes in dataset, SPD/DI not applicable. Absence of PII documented. Explainability: SHAP (DeepExplainer): pH and Sulfate identified as dominant features. LIME local explanations for potable and non-potable samples,. Robustness: Threshold tuning across 0.5-0.8, input validation for impossible values Security: Input sanitisation, rate limiting, security headers, prediction logging, real-world incident documented		Recall on unsafe water at or above 0.90, False negatives below 30 per 656 test samples Both models return predictions within 2 seconds API rejects invalid inputs with descriptive error messages All predictions logged for audit Human review documented as required before action Model card and data card approved and archived GitHub repository publicly accessible for inspection	
END OF LIFE PLAN			
Retirement triggers: Azure student credits exhausted or course submission deadline passed. Retirement steps: Stop Azure VM, download prediction logs, archive model files on GitHub with retirement tag, update model card with retirement reason and date. GitHub repository will remain publicly accessible after VM shutdown. This is a course project deployment, no formal production transition plan applies.			
APPROVAL GATE		RESIDUAL RISK	
Approved for: Field screening of water samples Not approved for: Final safety certification, automated decisions without human review, biological contamination detection		Accepted residual risk: 19 false negatives per 656 samples at threshold 0.7.Risk owner acknowledges that no physicochemical model can detect biological threats. Human review required for all predictions before action.	